

Algorithm for file updates in Python

Project description

In this project, I use Python to update an `allow_list` of IP addresses. I read the IP addresses from a file and compare them with a list of addresses that need to be removed. I use a loop and conditional statement to remove the unwanted IP addresses from the allow list. Finally, I update the file with the revised list of allowed IP addresses.

Open the file that contains the allow list

```
import_file = "allow_list.txt"
```

With `open(import_file, "r")` as file:

First, I assign "allow_list.txt" to the `import_file` variable. Then, I use a "with" statement and the `open()` function to open the file in read mode. I use "as file" to store the file so I can work with it inside the with statement.

Read the file contents

```
ip_addresses = file.read()
```

I use the `.read()` method to read the contents of the allow list file and convert them into a string. I store the string in a variable called `ip_addresses`.

Convert the string into a list

```
ip_addresses = ip_addresses.split()
```

I use the `.split()` method on the `ip_addresses` variable to convert the string into a list. This allows me to work with each IP address individually and remove the ones I need to.

Iterate through the remove list

```
for element in remove_list:
```

I use a for loop to go through each IP address in the `remove_list`. I use "element" as the loop variable, which represents each IP address as the loop goes through the list.

Remove IP addresses that are on the remove list

```
if element in remove_list:
```

```
    ip_addresses.remove(element)
```

I use an “if” statement to check whether “element” is in the remove_list. If it is, I use the .remove() method to remove that IP address from the list.

Update the file with the revised list of IP addresses

```
ip_addresses = "\n".join(ip_addresses)
```

```
with open(import_file, "w") as file:
```

```
    file.write(ip_addresses)
```

Summary

I start by opening the allow list file and reading the IP addresses into a string. I convert the string into a list so I can work with each IP address separately. I use a for loop and an if statement to find and remove the IP addresses that are on the remove list. Then, I use the .join() method to convert the updated list back into a string with each IP address on a new line. Finally, I use a with statement and the .write() method to update the allow list file with the revised IP addresses.